

EXPERIMENT NO 11

Determination and plotting of Magnetic Flux density in Ferromagnetic Materials with different (i) length, (ii) width and (iii) magnetic field intensity.

Aim:

To write program for Magnetic flux density in ferromagnetic materials using SCILAB.

Software required:

- SCILAB version 6.0.1

Program:

```
EXP 11.sci (C:\Users\user\OneDrive\Documents\EXP 11.sci) - SciNotes
File Edit Format Options Window Execute ?
EXP 11.sci (C:\Users\user\OneDrive\Documents\EXP 11.sci) - SciNotes
EXP 11.sci X
1  clc;
2  clear;
3
4  // Input values
5  mr = input("Enter the value of relative permeability (mr) :- ");
6  H = input("Enter the value of magnetic field intensity (H) :- ");
7  L = input("Enter the value of Length (m) :- ");
8  W = input("Enter the value of Width (m) :- ");
9
10 // Constants
11 mu0 = 4 * pi * 10^-7;
12
13 // Area
14 A = L * W;
15
16 // Magnetic flux density
17 B = mu0 * mr * H;
18
19 // Magnetic flux
20 Phi = B * A;
21
22 // Display results
23 disp("Area :- ", A);
24 disp("Magnetic flux density B :- ", B, " T");
25 disp("Magnetic flux Phi :- ", Phi, " Wb");
26
27 // Plot
28 x = linspace(0, 10, 50);
29 y = B * exp(-0.1 * x);
30
31 clf(0);
32 plot(x, y);
33 xlabel("Distance");
34 ylabel("Magnetic Flux Density B (T)");
35 title("Magnetic Flux Density");
36 grid();
37
```

Output:

Scilab 2026.1.0 Console

File Edit Control Applications ?

File Browser

C:\Users\user\Documents\

Enter the value of relative permeability (μ_r) = 3×10^{-7}

Enter the value of magnetic field intensity (H) = 6

Enter the value of Length (m) = 4

Enter the value of Width (m) = 2

"Area = "

8.

"Magnetic flux density B = "

0.0001734

" T"

"Magnetic flux Phi = "

0.0013873

" Wb"

-->

Variable Browser

Name	Value	Type	Visibility	Memory
A	8	Double	local	216 B
B	0.000173	Double	local	216 B
H	6	Double	local	216 B
L	4	Double	local	216 B
Phi	0.00139	Double	local	216 B
W	2	Double	local	216 B
ans	1x1 Graphic ...	Graphic	local	216 B
mr	23	Double	local	216 B
mu0	1.26e-06	Double	local	216 B

Command History

-- 10/08/2026 13:50:17 -- //

-- 10/08/2026 14:33:01 -- //

-- 10/08/2026 14:33:17 -- //

-- 10/08/2026 14:50:34 -- //

-- 24/08/2026 04:27:36 -- //

-- 24/08/2026 04:29:27 -- //

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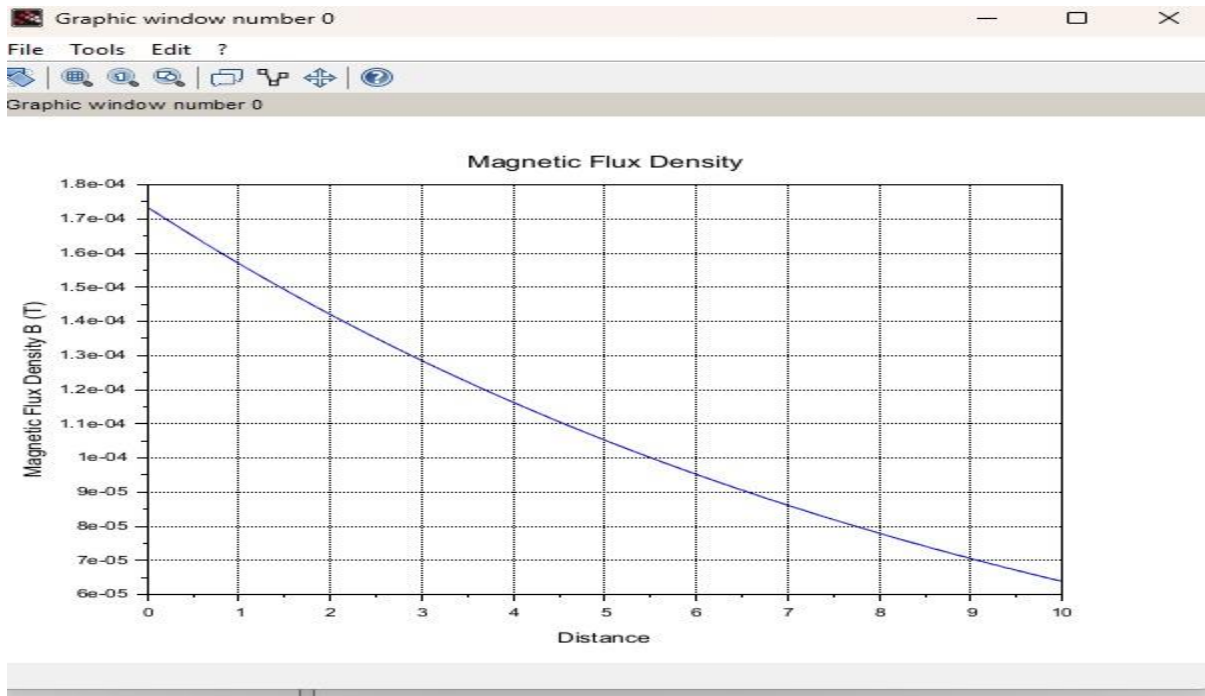
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Theoretical calculations:

EXP NO : 11

Determine and plotting of Magnetic Flux density in Ferromagnetic Materials with different (i) length, (ii) width and (iii) magnetic field intensity.

Output :

Enter the value of $\mu_r = 3 \times 10^7$

Enter the value of $H = 6 \text{ mm}$

Enter the value of length = 4 mm

Enter the value of width = 2 mm

Practical

Practical Output :

" Area = 8 mm^2

" Magnetic flux density $B = 0.0001734 \text{ T}$

" Magnetic flux $\phi = 0.00013873 \text{ Wb}$

Theoretical Calculation :

Given,

$\mu_r = 3 \times 10^7$, $H = 6 \text{ A/m}$, $l = 4 \text{ mm}$, $w = 2 \text{ mm}$

Area :

$$A = l \times w = 4 \times 2 = 8 \text{ mm}^2 = 8 \times 10^{-6} \text{ m}^2$$

Magnetic flux density :

$$B = \mu_0 \mu_r H$$

$$B = (4\pi \times 10^{-7}) (3 \times 10^7) (6)$$

$$B = 22.62 \text{ T}$$

Magnetic flux :

$$\phi = BA$$

$$\phi = 22.62 (8 \times 10^{-6})$$

$$\phi = 1.81 \times 10^{-4} \text{ Wb}$$

Case 1: Examine the effect of length of a ferromagnetic material on the magnetic flux density (B).

Analytical :

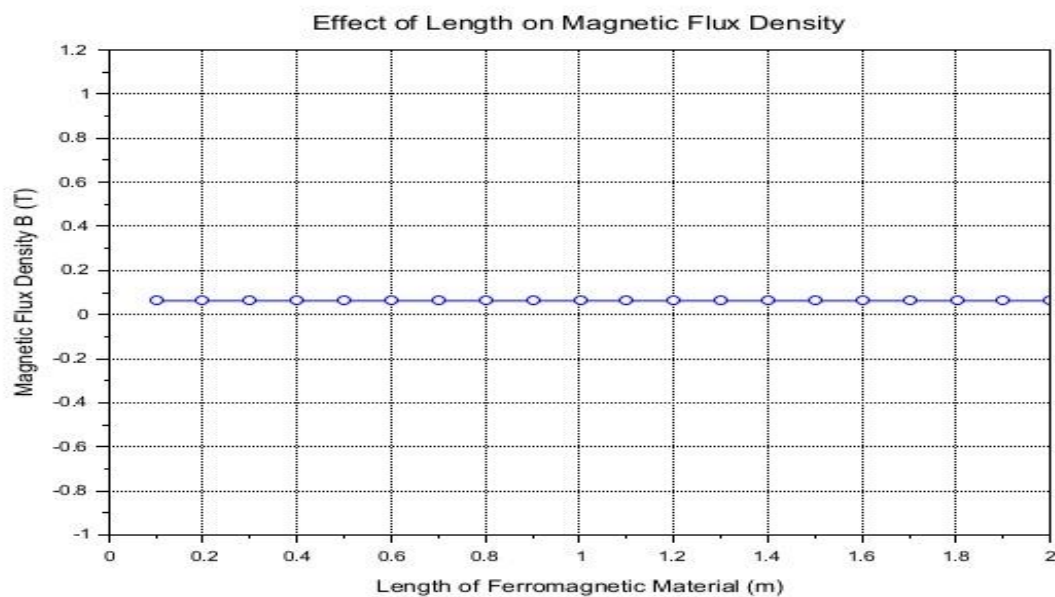
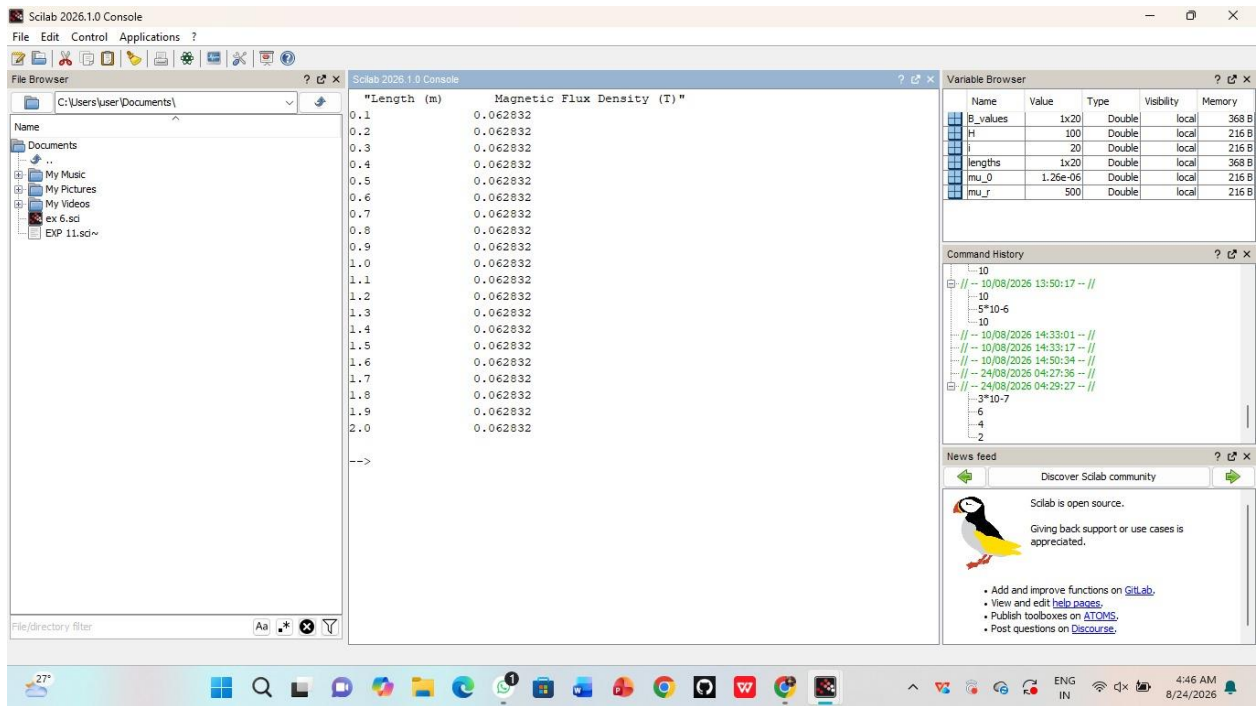
Calculate the magnetic flux density for the following conditions

- The material is ferromagnetic with relative permeability (μ_r) of 500.
- The magnetic field intensity (H) is held constant at 100 A/m.
- The length of the material varies from 0.1 m to 2 m.

Program:

```
EXP 11.sci (C:\Users\user\OneDrive\Documents\EXP 11.sci) - SciNotes
File Edit Format Options Window Execute ?
EXP 11.sci (C:\Users\user\OneDrive\Documents\EXP 11.sci) - SciNotes
EXP 11.sci X
1 %lc;
2 clear;
3
4 //Constants
5 mu_r = 500; //Relative permeability
6 H = 100; //Magnetic field intensity (A/m)
7 lengths = 0.1:0.1:2; //Lengths from 0.1 m to 2 m
8 mu_0 = 4 * pi * 10^(-7);
9
10 //Initialise array
11 B_values = zeros(1, length(lengths));
12
13 //Calculate magnetic flux density
14 for i = 1:length(lengths)
15     B_values(i) = mu_r * mu_0 * H;
16 end
17
18 //Display results
19 disp('Length (m) ----- Magnetic Flux Density (T)');
20
21 for i = 1:length(lengths)
22     fprintf('%1f ----- %1f\n', lengths(i), B_values(i));
23 end
24
25 //Plot
26 clf;
27 plot(lengths, B_values, '-o');
28
29 xlabel('Length of Ferromagnetic Material (m)');
30 ylabel('Magnetic Flux Density B (T)');
31 title('Effect of Length on Magnetic Flux Density');
32
33 xgrid();
34
```

Output:



Case 2: To investigate how the magnetic flux density (B) in a ferromagnetic material changes with

Different widths at a constant length and magnetic field intensity.

Analytical:

Calculate magnetic flux density (B) in a ferromagnetic material changes with different widths at a constant length

And magnetic field intensity for the following conditions

- Material: Soft iron (same as in Case Study 1)
- Length (L): Constant at 10 cm
- Magnetic Field Intensity (H): Constant at 1000 A/m
- Width (W): Varies from 2 cm to 10 cm (i.e., 2 cm, 4 cm, 6 cm, 8 cm, 10 cm)

Program:

```
EXP 11.sci (C:\Users\user\OneDrive\Documents\EXP 11.sci) - SciNotes
File Edit Format Options Window Execute ?

EXP 11.sci (C:\Users\user\OneDrive\Documents\EXP 11.sci) - SciNotes
EXP 11.sci x

1  clc;
2  clear;
3
4  //Constants
5  L = 0.1; //Length in meters (10 cm)
6  H = 1000; //Magnetic field intensity in A/m
7  mu_0 = 4 * pi * 10^(-7); //Permeability of free space (H/m)
8
9  //Widths in meters (2 cm to 10 cm)
10 widths = [0.02, 0.04, 0.06, 0.08, 0.10];
11 num_widths = length(widths);
12
13 //Relative permeability of soft iron
14 mu_r = 1000;
15
16 //Initialize array
17 B_values = zeros(1, num_widths);
18
19 //Calculate magnetic flux density
20 for i = 1:num_widths
21     B_values(i) = mu_0 * mu_r * H;
22 end
23
24 //Display results
25 disp("Width (m) --- Flux Density (T)");
26
27 for i = 1:num_widths
28     fprintf("%5.2f-----%.6f\n", widths(i), B_values(i));
29 end
30
31 //Plot the results
32 clf;
33 plot(widths, B_values, "-o");
34 xlabel("Width of Ferromagnetic Material (m)");
35 ylabel("Magnetic Flux Density B (T)");
36 title("Effect of Width on Magnetic Flux Density");
37 grid();
38
```

Output:

Scilab 2026.1.0 Console

File Edit Control Applications ?

File Browser

C:\Users\user\Documents\

Name

- Documents
- My Music
- My Pictures
- My Videos
- ex 6.sci
- EXP 11.sci

File/Directory filter

Scilab 2026.1.0 Console

```
"Width (m)    Flux Density (T)"
0.02          1.256637
0.04          1.256637
0.06          1.256637
0.08          1.256637
0.10          1.256637
-->
```

Variable Browser

Name	Value	Type	Visibility	Memory
B_values	1x5	Double	local	248 B
H	1e+03	Double	local	216 B
L	0.1	Double	local	216 B
mu_0	1.256e-06	Double	local	216 B
mu_r	1e+03	Double	local	216 B
num_widths	5	Double	local	216 B
widths	1x5	Double	local	248 B

Command History

```
// -- 10/08/2026 13:50:17 -- //
10
5*10^-6
10
// -- 10/08/2026 14:33:01 -- //
// -- 10/08/2026 14:33:17 -- //
// -- 10/08/2026 14:50:34 -- //
// -- 24/08/2026 04:27:36 -- //
// -- 24/08/2026 04:29:27 -- //
3*10^-7
6
4
2
```

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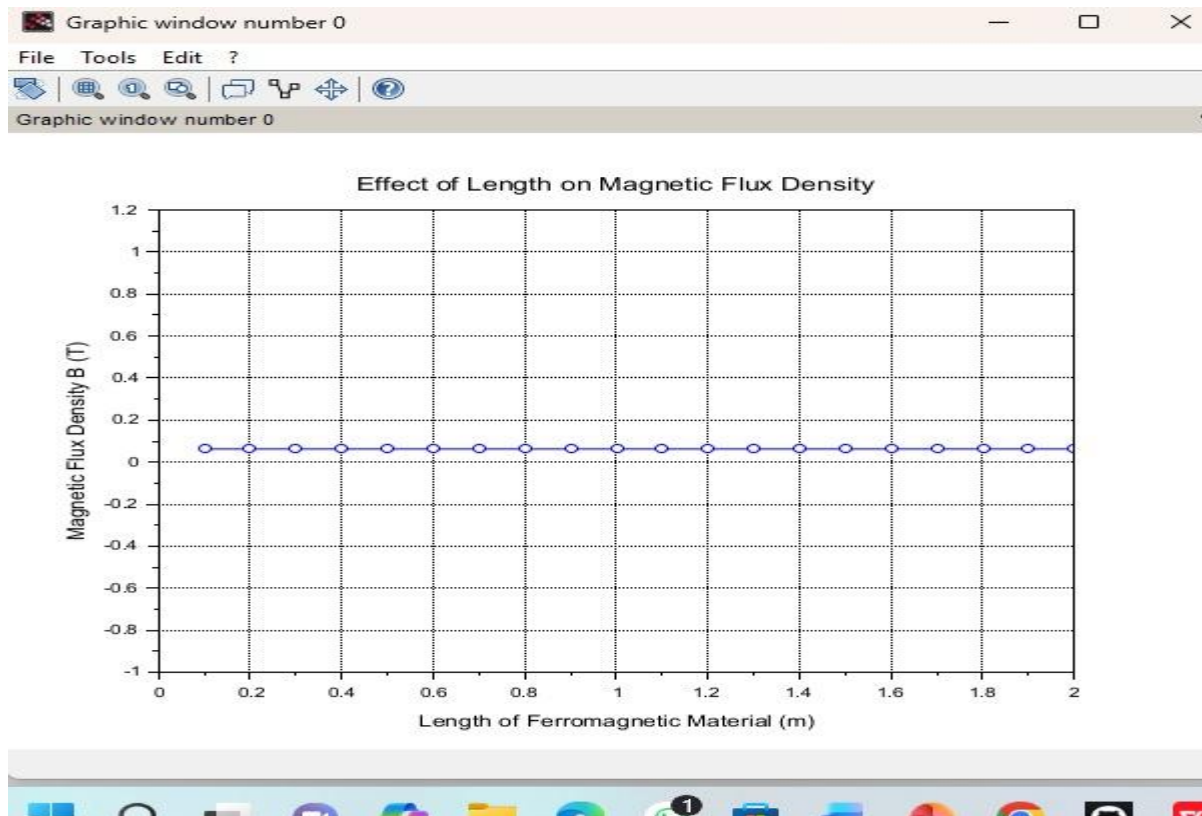
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Case 3: To examine how the magnetic flux density (B) in a ferromagnetic material change with different values of magnetic field intensity at constant length and width.

Program:

```
EXP 11.sci (C:\Users\user\OneDrive\Documents\EXP 11.sci) - SciNotes
File Edit Format Options Window Execute ?
[Icons]
EXP 11.sci (C:\Users\user\OneDrive\Documents\EXP 11.sci) - SciNotes
EXP 11.sci X
3
4 //Define constants
5 L = 0.1; //Length in meters (10 cm)
6 W = 0.05; //Width in meters (5 cm)
7
8 H_values = [200, 500, 1000, 1500, 2000]; //H in A/m
9
10 mu_0 = 4 * pi * 10^(-7); //Permeability of free space (H/m)
11 mu_r_initial = 1000; //Relative permeability of soft iron
12
13 //Initialise array
14 B_values = zeros(1, length(H_values));
15
16 //Calculate B for each H value
17 for i = 1:length(H_values)
18     H = H_values(i);
19     mu_r = mu_r_initial;
20     //Calculate absolute permeability
21     mu = mu_0 * mu_r;
22     //Calculate magnetic flux density
23     B_values(i) = mu * H;
24 end
25
26 //Display results
27 disp("Magnetic Field Intensity (A/m) --- Magnetic Flux Density (T)");
28
29 for i = 1:length(H_values)
30     fprintf("%10.0f-----%.2f\n", H_values(i), B_values(i));
31 end
32
33 //Plotting the results
34 clf();
35 plot(H_values, B_values, "-o");
36
37 xlabel("Magnetic Field Intensity (A/m)");
38 ylabel("Magnetic Flux Density (T)");
39 title("Magnetic Flux Density vs Magnetic Field Intensity for Soft Iron");
40
41
42
43 grid();
```

Output:

Scilab 2026.1.0 Console

File Edit Control Applications ?

File Browser: C:\Users\User\Documents

Scilab 2026.1.0 Console

```

Magnetic Field Intensity (A/m)   Magnetic Flux Density (T) *
200                               0.251327
500                               0.628319
1000                              1.256637
1500                              1.884956
2000                              2.513274
--> |

```

Variable Browser

Name	Value	Type	Visibility	Memory
B_values	1x5	Double	local	248 B
H	2e+03	Double	local	216 B
H_values	1x5	Double	local	248 B
L	0.1	Double	local	216 B
W	0.05	Double	local	216 B
r	5	Double	local	216 B
mu	0.00126	Double	local	216 B
mu_0	1.26e-06	Double	local	216 B
mu_r	1e+03	Double	local	216 B

Command History

```

-- 10
-- // -- 10/08/2026 13:50:17 -- //
-- 10
-- 5*10-6
-- 10
-- // -- 10/08/2026 14:33:01 -- //
-- // -- 10/08/2026 14:33:17 -- //
-- // -- 10/08/2026 14:50:34 -- //
-- // -- 24/08/2026 04:27:36 -- //
-- // -- 24/08/2026 04:29:27 -- //
-- 3*10-7
-- 6
-- 4
-- 2

```

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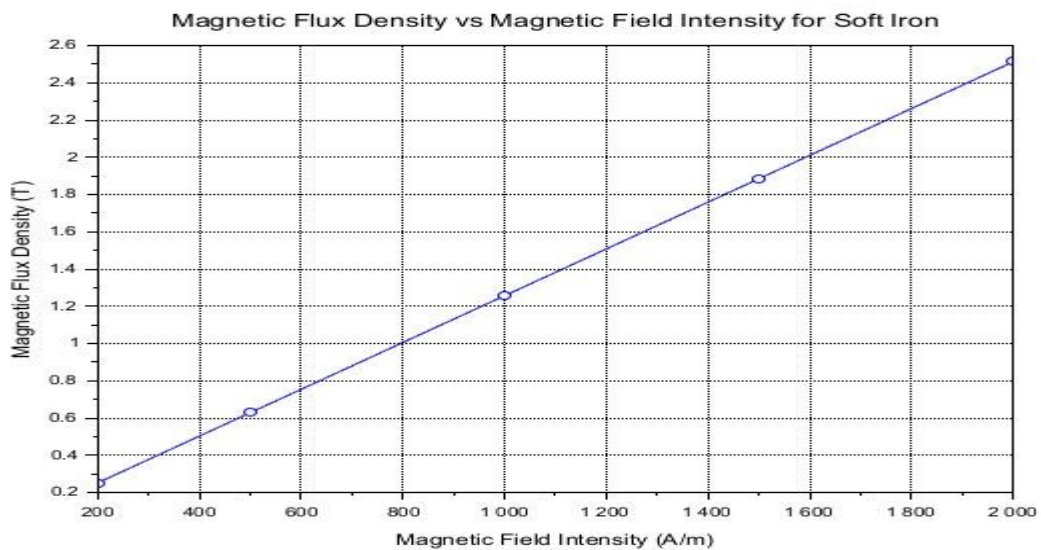
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Graphic window number 0

File Tools Edit ?

Graphic window number 0



Result:

The effect of Magnetic field intensity of a ferromagnetic material on the magnetic flux density (B) is examined.